

DIY ARCADE CABINET BUILD GUIDE

Youth-Center Staff Guide for Legal Custom Stations

Parts • Legal Games • Build Day • Daily Routine • Ownership

Prepared for Coin Op 4 Charity / Arcade Angels

Build a real-feeling arcade station from cheap parts, run only legal games, and keep it safe for a youth-center floor.

1. Why Build Your Own

A youth center does not need a store-bought home cabinet to put a real arcade on the floor. A custom station sidesteps consumer licensing terms that limit public use, costs far less than a licensed upright, and lets the kids own the look and the game list. That ownership is the point: a kid who painted the marquee and fixed the stick is not going to trash the next cabinet.

- Zero consumer-cabinet license risk — no home-only terms to violate on a club floor.
- Kids design the art and pick the games, so the machine is their project.
- Parts cost roughly \$50–\$150 per station versus several hundred for a licensed home cabinet.
- Pairs with Coin Op 4 Charity tablet terminals and restored vintage machines already in the fleet.

2. What You Are Building

A free-play upright or bartop made of four parts: a small computer, a screen, arcade controls, and a case. Software boots into a locked menu. Games come only from legal sources listed in section 4. This is a complement to restored original cabinets and to the tablet + BSP-Y02 stations already documented in this folder — not a replacement for either.

3. Parts List (Per Station)

Part	Example	Approx. cost
Computer	Raspberry Pi 5 (4 GB) or used mini PC	\$60–\$120
Storage	32 GB microSD or small SSD	\$8–\$15
Screen	15–19" HDMI monitor or donated TV	\$0–\$60
Controls	USB stick or \$9–\$17 joystick + buttons	\$10–\$25
Case	3D-printed shell, scrap wood, thrifted cab	\$0–\$40
Power	Surge strip, wall mount, cable ties	\$10

Total target: under \$150 if the screen and case are donated. Do not cheap out on the surge strip.

4. Software — Legal by Default

Emulators and front-ends are legal tools. The game files are the part that is not, unless you have a lawful source. Staff should treat this as a hard rule, not a suggestion.

- Flash Batocera or RetroArch onto the storage card. Boot into a locked kiosk / attract menu so volunteers never hunt settings.
- Original carts and discs the club or kids already own, dumped with a legal hardware tool the organization controls.
- Homebrew and original games the kids make themselves.
- Games submitted through Arrrcadé OS on coinop4charity.org after review for safety and fit.
- No downloaded commercial ROM packs. Ever. If a file did not come from one of the three sources above, it does not go on the machine.

Kiosk page used on the live site for tablet stations: <https://coinop4charity.org/?kiosk=1> — the same terminal can sit on a Pi-driven cabinet if you prefer a browser frontend over a local emulator list.

5. Build Day With the Kids

Treat the first cabinet as a workshop, not a drop-off. Staff handle power, wiring, and the kiosk lock. Kids handle the look and the first game list.

- Weekend or after-school block: kids design marquee art and side graphics (markers, vinyl, 3D-printed badges).
- They pick the first ten legal titles and write the names on a card taped inside the coin door or back panel.
- One adult signs off that every game file has a documented source.
- Name the station. A named machine gets treated better than “the arcade.”

6. Safety & Placement Checklist

- ☐ All cables secured; no trip hazards; one surge-protected strip for the whole station.
- ☐ Dedicated wall outlet — no daisy-chained cheap strips.
- ☐ Screen at kid height on a stable base; cabinet cannot tip if leaned on.
- ☐ Drinks stay off the control panel. Wipe controls on a regular hygiene schedule.
- ☐ Kiosk / staff PIN known only to adults.
- ☐ Kids named as the maintenance crew for that cabinet — they own it.

7. Daily Staff Routine

Morning (or start of program):

- Power strip ON. Wait for the menu / terminal to appear.
- Wiggle the stick and press A / Start on a known game.
- Note any dead button, dim screen, or odd noise on a one-line log.

Closing:

- Power the strip off unless the room is locked and you prefer to leave it charging.
- Log broken parts the same day so the kid crew can fix them before the next session.

8. Staff Card (Copy for the Cabinet)

- Play is free.
- Menu should load by itself after power-on.
- A or Start = play. Select = back. Hold Start = exit a game.
- B does not quit games.
- If the screen is wrong or frozen, ask staff. Settings and PINs are for adults only.
- Tell staff if a button sticks or the stick drifts.

9. When Something Breaks

- Stick dead — reseal USB; if wireless, charge and press HOME once after the first pairing. Do not hold obscure combo sequences for daily use.
- Black screen — HDMI cable and power strip first. Do not reflash the card until those two are ruled out.
- Game will not launch — that is a file problem. Swap the title. Do not reflash the whole system for one bad game.
- Cabinet rattle or tip — tighten the base before the next open hours. Do not leave an unstable box on the floor.

10. Notes for Host Sites

This station is a low-cost free-play complement to full restored cabinets and to the tablet + BSP-Y02 stations. It is meant to be interesting, shareable, and maintainable — not a locked tournament machine and not a gray-market multicade. When power and wake steps stay this short, staff protect program time for young people instead of fighting devices.

Contact Coin Op 4 Charity / Arcade Angels for placement support, spare sticks, a parts review, or to place a finished station. Open an issue on the public repo or use the form at coinop4charity.org.

Related docs in the same set: [Tablet_Arcade_Station_Youth_Center_Setup_Guide.pdf](#) and [BSP-Y02_Web_Arcade_Controller_Guide.pdf](#).

11. Document Control

Guide type: DIY youth-center cabinet • Hardware: Raspberry Pi / mini PC + HDMI screen + USB or wired arcade controls • Software: Batocera / RetroArch or coinop4charity.org/?kiosk=1 • Audience: site staff and volunteers • Companion to restoration and controller guides under docs/restoration/.